

Experiment 1: Exploratory Data Analysis

Spambase Dataset

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1 Aim

To explore and understand the functionalities of core Python libraries used in Machine Learning – NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib/Seaborn – and to apply exploratory data analysis (EDA) techniques (data cleaning, outlier removal, visualization, correlation analysis) on the Spambase dataset to identify the type of ML task and suitable algorithms.

2 Exploratory Data Analysis: Spambase Dataset

2.1 Console Output

The script below was run on spambase csv.csv. The full terminal output is reproduced here for reference.

```
~/ML-EXP1 >> python3 eda.py
=====
EXPLORATORY DATA ANALYSIS
=====

First 5 Rows
  word_freq_make  word_freq_address  ...  capital_run_length_total  class
0           0.00           0.64  ...              278              1
1           0.21           0.28  ...             1028              1
2           0.06           0.00  ...             2259              1
3           0.00           0.00  ...              191              1
4           0.00           0.00  ...              191              1

[5 rows x 58 columns]

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 4601 entries, 0 to 4600
Data columns (total 58 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   word_freq_make                        4601 non-null   float64
1   word_freq_address                    4601 non-null   float64
2   word_freq_all                        4601 non-null   float64
3   word_freq_3d                        4601 non-null   float64
4   word_freq_our                        4601 non-null   float64
5   word_freq_over                       4601 non-null   float64
6   word_freq_remove                     4601 non-null   float64
7   word_freq_internet                   4601 non-null   float64
8   word_freq_order                      4601 non-null   float64
9   word_freq_mail                       4601 non-null   float64
10  word_freq_receive                    4601 non-null   float64
11  word_freq_will                       4601 non-null   float64
12  word_freq_people                     4601 non-null   float64
13  word_freq_report                     4601 non-null   float64
14  word_freq_addresses                  4601 non-null   float64
15  word_freq_free                       4601 non-null   float64
```

16	word_freq_business	4601 non-null	float64
17	word_freq_email	4601 non-null	float64
18	word_freq_you	4601 non-null	float64
19	word_freq_credit	4601 non-null	float64
20	word_freq_your	4601 non-null	float64
21	word_freq_font	4601 non-null	float64
22	word_freq_000	4601 non-null	float64
23	word_freq_money	4601 non-null	float64
24	word_freq_hp	4601 non-null	float64
25	word_freq_hpl	4601 non-null	float64
26	word_freq_george	4601 non-null	float64
27	word_freq_650	4601 non-null	float64
28	word_freq_lab	4601 non-null	float64
29	word_freq_labs	4601 non-null	float64
30	word_freq_telnet	4601 non-null	float64
31	word_freq_857	4601 non-null	float64
32	word_freq_data	4601 non-null	float64
33	word_freq_415	4601 non-null	float64
34	word_freq_85	4601 non-null	float64
35	word_freq_technology	4601 non-null	float64
36	word_freq_1999	4601 non-null	float64
37	word_freq_parts	4601 non-null	float64
38	word_freq_pm	4601 non-null	float64
39	word_freq_direct	4601 non-null	float64
40	word_freq_cs	4601 non-null	float64
41	word_freq_meeting	4601 non-null	float64
42	word_freq_original	4601 non-null	float64
43	word_freq_project	4601 non-null	float64
44	word_freq_re	4601 non-null	float64
45	word_freq_edu	4601 non-null	float64
46	word_freq_table	4601 non-null	float64
47	word_freq_conference	4601 non-null	float64
48	char_freq_%3B	4601 non-null	float64
49	char_freq_%28	4601 non-null	float64
50	char_freq_%5B	4601 non-null	float64
51	char_freq_%21	4601 non-null	float64
52	char_freq_%24	4601 non-null	float64
53	char_freq_%23	4601 non-null	float64
54	capital_run_length_average	4601 non-null	float64
55	capital_run_length_longest	4601 non-null	int64
56	capital_run_length_total	4601 non-null	int64
57	class	4601 non-null	int64

dtypes: float64(55), int64(3)

memory usage: 2.0 MB

Dataset Shape: (4601, 58)

Column Names

['word_freq_make', 'word_freq_address', 'word_freq_all', 'word_freq_3d', 'word_freq_our', 'word_freq_over', 'word_freq_remove', 'word_freq_internet', 'word_freq_order', 'word_freq_mail', 'word_freq_receive', 'word_freq_will', 'word_freq_people', 'word_freq_report', 'word_freq_addresses', 'word_freq_free', 'word_freq_business', 'word_freq_email', 'word_freq_you', 'word_freq_credit', 'word_freq_your', 'word_freq_font', 'word_freq_000', 'word_freq_money', 'word_freq_hp', 'word_freq_hpl', 'word_freq_george', 'word_freq_650', 'word_freq_lab', 'word_freq_labs', 'word_freq_telnet', 'word_freq_857', 'word_freq_data', 'word_freq_415', 'word_freq_85', 'word_freq_technology', 'word_freq_1999', 'word_freq_parts', 'word_freq_pm', 'word_freq_direct', 'word_freq_cs', 'word_freq_meeting', 'word_freq_original', 'word_freq_project', 'word_freq_re', 'word_freq_edu', 'word_freq_table', 'word_freq_conference', 'char_freq_%3B', 'char_freq_%28', 'char_freq_%5B', 'char_freq_%21', 'char_freq_%24', 'char_freq_%23', 'capital_run_length_average', 'capital_run_length_longest', 'capital_run_length_total', 'class']

Missing Values Before Cleaning

[all 58 columns show 0 missing values]

Missing Values After Cleaning

[all 58 columns show 0 missing values]

Duplicate Rows: 391

Shape After Removing Duplicates: (4210, 58)

Data Types

[55 columns: float64, capital_run_length_longest: int64, capital_run_length_total: int64, class: int64]

Unique Values Per Column

word_freq_make	142
word_freq_address	171
word_freq_all	214
word_freq_3d	43
word_freq_our	255
word_freq_over	141
word_freq_remove	173
word_freq_internet	170
word_freq_order	144
word_freq_mail	245
word_freq_receive	113
word_freq_will	316
word_freq_people	158
word_freq_report	133
word_freq_addresses	118
word_freq_free	253
word_freq_business	197
word_freq_email	229
word_freq_you	575
word_freq_credit	148
word_freq_your	401
word_freq_font	99
word_freq_000	164
word_freq_money	143
word_freq_hp	395
word_freq_hpl	281
word_freq_george	240
word_freq_650	200
word_freq_lab	156
word_freq_labs	179
word_freq_telnet	128
word_freq_857	106
word_freq_data	184
word_freq_415	110
word_freq_85	177
word_freq_technology	159
word_freq_1999	188
word_freq_parts	53
word_freq_pm	163
word_freq_direct	125
word_freq_cs	108
word_freq_meeting	186
word_freq_original	136
word_freq_project	160
word_freq_re	230
word_freq_edu	227
word_freq_table	38
word_freq_conference	106
char_freq_%3B	313
char_freq_%28	641
char_freq_%5B	225
char_freq_%21	964
char_freq_%24	504
char_freq_%23	316
capital_run_length_average	2161
capital_run_length_longest	271
capital_run_length_total	919
class	2

Numerical Statistics

	word_freq_make	word_freq_address	...	capital_run_length_total	class
count	4210.000000	4210.000000	...	4210.000000	4210.000000
mean	0.104366	0.112656	...	291.181948	0.398812
std	0.300005	0.454260	...	618.654838	0.489712
min	0.000000	0.000000	...	1.000000	0.000000
25%	0.000000	0.000000	...	40.000000	0.000000
50%	0.000000	0.000000	...	101.500000	0.000000
75%	0.000000	0.000000	...	273.750000	1.000000
max	4.540000	14.280000	...	15841.000000	1.000000

[8 rows x 58 columns]

Numeric Columns (features only)

```
['word_freq_make', 'word_freq_address', 'word_freq_all', 'word_freq_3d', 'word_freq_our', 'word_freq_over',  
'word_freq_remove', 'word_freq_internet', 'word_freq_order', 'word_freq_mail', 'word_freq_receive',  
'word_freq_will', 'word_freq_people', 'word_freq_report', 'word_freq_addresses', 'word_freq_free',  
'word_freq_business', 'word_freq_email', 'word_freq_you', 'word_freq_credit', 'word_freq_your', 'word_freq_font',  
'word_freq_000', 'word_freq_money', 'word_freq_hp', 'word_freq_hpl', 'word_freq_george', 'word_freq_650',  
'word_freq_lab', 'word_freq_labs', 'word_freq_telnet', 'word_freq_857', 'word_freq_data', 'word_freq_415',  
'word_freq_85', 'word_freq_technology', 'word_freq_1999', 'word_freq_parts', 'word_freq_pm', 'word_freq_direct',  
'word_freq_cs', 'word_freq_meeting', 'word_freq_original', 'word_freq_project', 'word_freq_re', 'word_freq_edu',  
'word_freq_table', 'word_freq_conference', 'char_freq_%3B', 'char_freq_%28', 'char_freq_%5B', 'char_freq_%21',  
'char_freq_%24', 'char_freq_%23', 'capital_run_length_average', 'capital_run_length_longest', 'capital_run_length_total']
```

Categorical Columns (incl. target if set)

```
['class']
```

Removing Outliers (IQR Method)

```
word_freq_make: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_address: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_all: 293 outliers flagged  
word_freq_3d: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_our: 449 outliers flagged  
word_freq_over: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_remove: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_internet: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_order: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_mail: 748 outliers flagged  
word_freq_receive: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_will: 250 outliers flagged  
word_freq_people: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_report: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_addresses: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_free: 830 outliers flagged  
word_freq_business: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_email: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_you: 58 outliers flagged  
word_freq_credit: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_your: 193 outliers flagged  
word_freq_font: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_000: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_money: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_hp: 1012 outliers flagged  
word_freq_hpl: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_george: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_650: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_lab: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_labs: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_telnet: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_857: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_data: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_415: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_85: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_technology: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_1999: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_parts: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_pm: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_direct: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_cs: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_meeting: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_original: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_project: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_re: 805 outliers flagged  
word_freq_edu: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_table: skipped (IQR is 0, likely a sparse/constant column)  
word_freq_conference: skipped (IQR is 0, likely a sparse/constant column)  
char_freq_%3B: skipped (IQR is 0, likely a sparse/constant column)  
char_freq_%28: 255 outliers flagged  
char_freq_%5B: skipped (IQR is 0, likely a sparse/constant column)  
char_freq_%21: 373 outliers flagged  
char_freq_%24: 730 outliers flagged  
char_freq_%23: skipped (IQR is 0, likely a sparse/constant column)
```

capital_run_length_average: 329 outliers flagged
capital_run_length_longest: 418 outliers flagged
capital_run_length_total: 497 outliers flagged

Shape Before Outlier Removal: (4210, 58)
Shape After Outlier Removal: (809, 58)

2.2 Visualizations

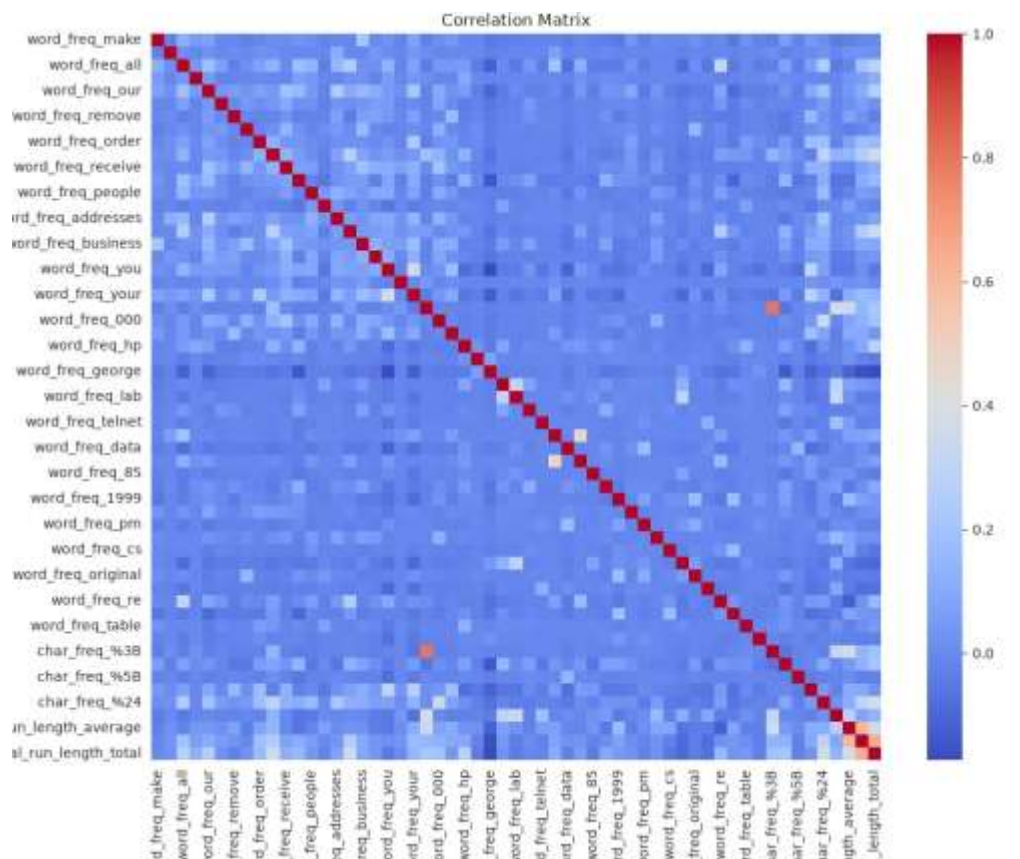


Figure 1: Correlation matrix of numeric features in the Spambase dataset. Most pairwise correlations are near zero, indicating limited multicollinearity among word-frequency features.

2.3 Summary Table

Table 1: Dataset shape at each cleaning stage

Stage	Rows	Columns
Original	4601	58
After missing value removal	4601	58
After duplicate removal	4210	58
After IQR outlier removal	809	58

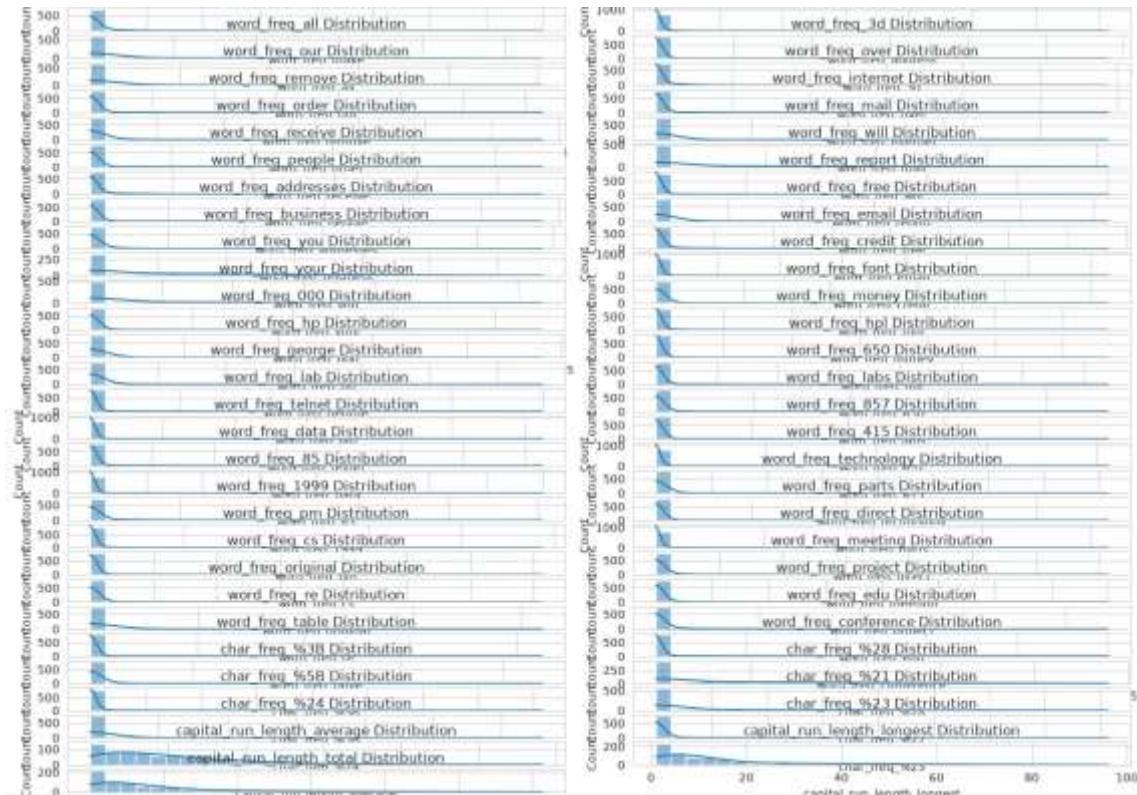


Figure 2: Distribution of numeric word_freq_* and char_freq_* features. Most features are zero-inflated and right-skewed, reflecting that individual words appear in only a small subset of emails.



Figure 3: Class distribution after cleaning and outlier removal (0 = non-spam, 1 = spam), showing a moderate class imbalance (616 vs. 193 samples).

2.4 Discussion

The numeric feature distributions (Figure 2) reveal that nearly all word freq_* columns are heavily zero-inflated and right-skewed: most individual words do not appear in most emails, so values are 0 for the majority of rows. This explains why many columns were skipped during IQR-based outlier removal ($Q1 = Q3 = 0$ for those columns).

The class distribution (Figure 3) shows a moderate class imbalance after cleaning: 616 non-spam versus 193 spam emails (roughly 76:24). This suggests that accuracy alone would be a misleading evaluation metric for a downstream spam classifier; precision, recall, F1-score, or ROC-AUC are more appropriate, along with possible class-weighting or resampling during model training.

Outlier removal reduced the dataset from 4210 rows (after deduplication) to 809 rows, an 81% reduction, driven by the compounding effect of applying IQR filtering across 34 skewed columns simultaneously. A milder filtering approach (e.g. a wider IQR multiplier, or skipping outlier removal on skewed, zero-inflated count-like features) would likely preserve more data for downstream classification.

3 Learning Outcomes

- Understood how to use NumPy and Pandas for efficient data handling and manipulation.
- Learned to identify and handle inconsistent missing-value representations in real-world data.
- Gained experience applying the IQR method for outlier detection and removal, and recognized its limitations on zero-inflated, skewed features.